CO1 - ASSESSMENT TOOL - 1

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1. Event communication security :-

a) Encrypt, "MEET AT MOON" using a caesar cipher with a shift of 4.

Plain Text : MEET AT NOON
Cipher Text : QIIX EX RSSR

Encrypted message : QIIX EX RSSR.

b). decryption :-
Shift backwards by 4 position.

eg:

* Q → M
* I → E
* X → T

2). Intercepted Suspicious Message :-

a) original message :
cipher text : GSRH RHZ HUXIVG
Plain text : THIS IS A SECRET

b). Justification :

* It is a encrypted using the Atbash cipher.
* In Atbash :
  * A ↔ Z
  * B ↔ Y
  * C ↔ X

3. Secure Messaging System (vigenere cipher):

Plain Text: HELLO
Key word: KEY

Repeat Keyword:-

plaintext: H E L L O
keyword: K E Y K E

Encryption:

H+K E+E L+Y L+K O+E
R I J V S

Cipher text: RIJVS.

4. Suspicious Digital Asset investigation:

a) detecting hidden data:-

Investigators can:

* use steganalysis tools
* Examine image pixels for unusual changes
* compare file size and metadata
* Perform statistical analysis.

b) Two security improvements:

1. encrypt the secret data
2. use randomized pixel position to hide the data.

5). Legacy encryption in corporate communication :-

a) Original message :-

Cipher Text : WE CRLTEERDSOEEFE AOCAIVDEN

Plain Text : WE ARE DISCOVERED FLEE AT ONCE

b) why it is vulnerable :-

* no secret key is used
* letter frequencies remain unchanged
* easy to break using brute - force
* Not suitable for modern secure communication.

c). Practical enhancement :-

* Replace the transposition cipher with a modern encryption algorithms such as Advanced encryption standard (AES)

* which provides strong confidentiality and resistance against attacks.